



CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. Diagrams may be drawn in pencil but **all writing** must be in **ink**.
5. All working **MUST** be **CLEARLY** shown.
6. The use of silent, non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalized.

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DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Acceleration due to gravity	g	=	9.80 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Boltzmann's constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Density of water		=	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water		=	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of ice		=	$3.34 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporization of water		=	$2.26 \times 10^6 \text{ J kg}^{-1}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$
Molar gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Stefan-Boltzmann constant	σ	=	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$

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SECTION A

Answer ALL questions in this section.

You MUST write your answers in the spaces provided in this booklet.

1. (a) Define the following terms and state which are vectors and which are scalar quantities.

(i) Displacement

(ii) Instantaneous velocity

(iii) Instantaneous acceleration

(iv) Kinetic energy

[6 marks]

- (b) Figure 1 shows data related to the displacement of a particle at various times, t . The displacement, x , of the particle is described by the equation,

$$x = x_0 \left(\frac{t}{t_0} \right)^n$$

where x_0 and t_0 are constants.

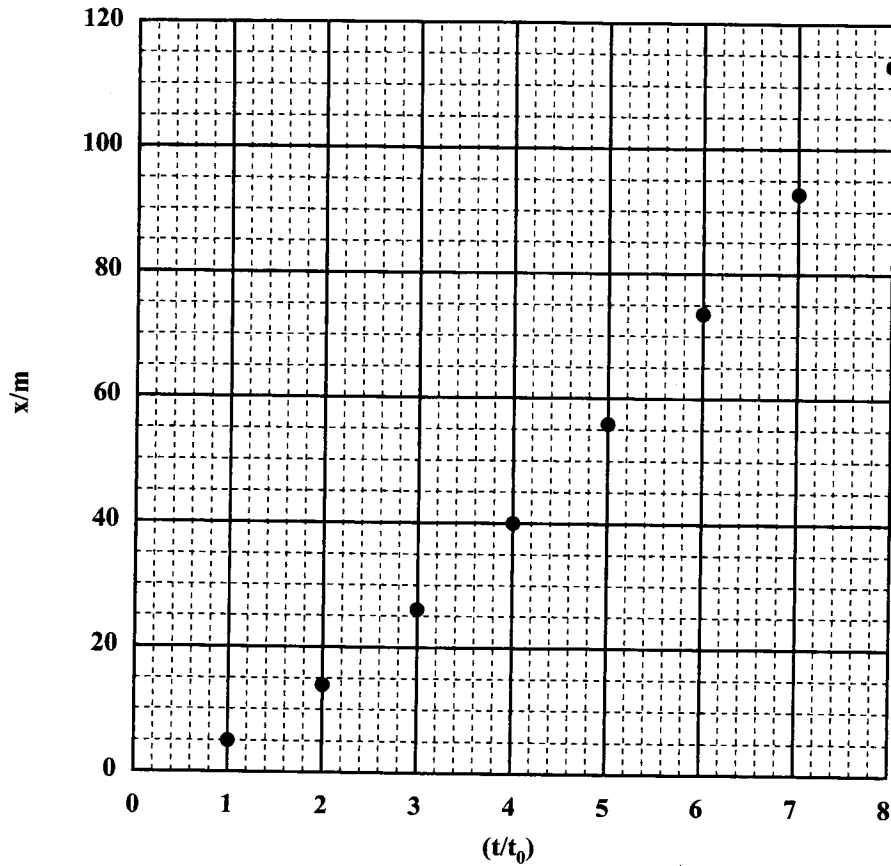


Figure 1

- (i) Read off the coordinates of any TWO points from the graph of Figure 1.

[2 marks]

(ii) Use the coordinates from (i) to calculate the value of

a) n

b) x_0 .

[3 marks]

[1 mark]

(iii) Given that $t_0 = 5$ s, determine the velocity of the particle when $t = 30$ s.

[3 marks]

Total 15 marks

2. (a) A mass, m , oscillates on the end of a coiled spring of spring constant, k . Write an expression for the period of oscillation, T , of the mass-spring system.

[1 mark]

- (b) Figure 2 illustrates a block which can be made to oscillate vertically between two identical springs, each with spring constant, k .

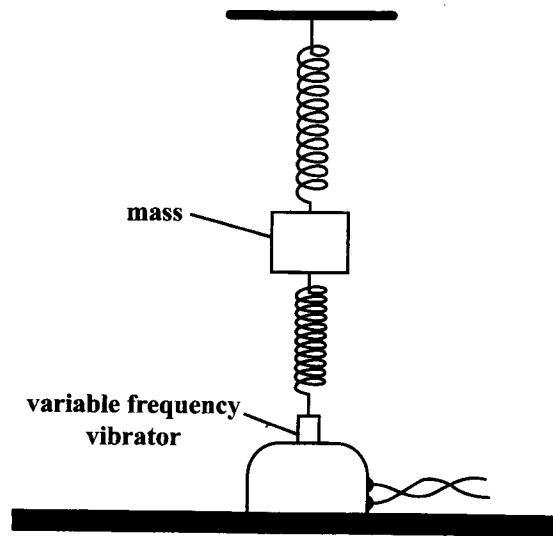
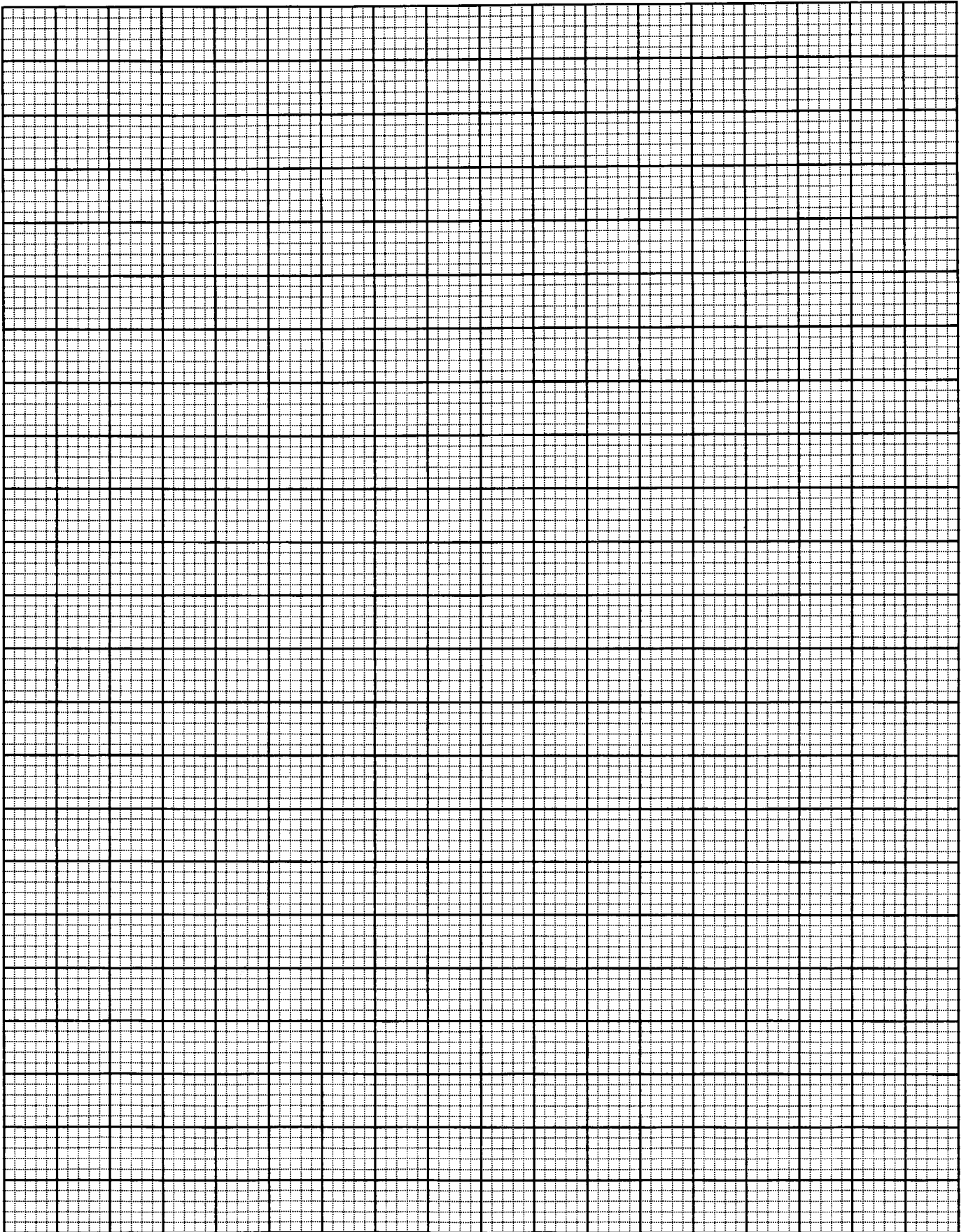


Figure 2

The vibrator has constant amplitude. As the frequency is changed, the amplitude of oscillation of the mass is measured. Table 1 shows the data recorded.

TABLE 1

Frequency, f/Hz	Amplitude of Vibration of Mass, y/cm
2.5	1.0
5.0	1.5
7.5	2.5
10	3.8
12.5	4.6
15	3.5
17.5	1.2
20	0.5



(i) On the grid provided on page 7, plot a graph of amplitude of vibration, y , against frequency, f . **[5 marks]**

(ii) For the block oscillating at maximum amplitude, determine the

a) angular frequency

[2 marks]

b) period of oscillation.

[1 mark]

(c) The mass of the block is 0.05 kg. Calculate the spring constant, k , for ONE of the springs.

[3 marks]

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- (d) (i) A small lump of plasticine is fixed to the mass. On your graph, draw a line to show the possible variation with frequency of the amplitude of the vibration of the combined mass.

[2 marks]

- (ii) Briefly describe ONE situation in which it is advantageous to use the phenomenon illustrated in this experiment.

[1 mark]

Total 15 marks

3. (a) Write an equation representing the First Law of Thermodynamics and state the meaning of EACH symbol used.

[2 marks]

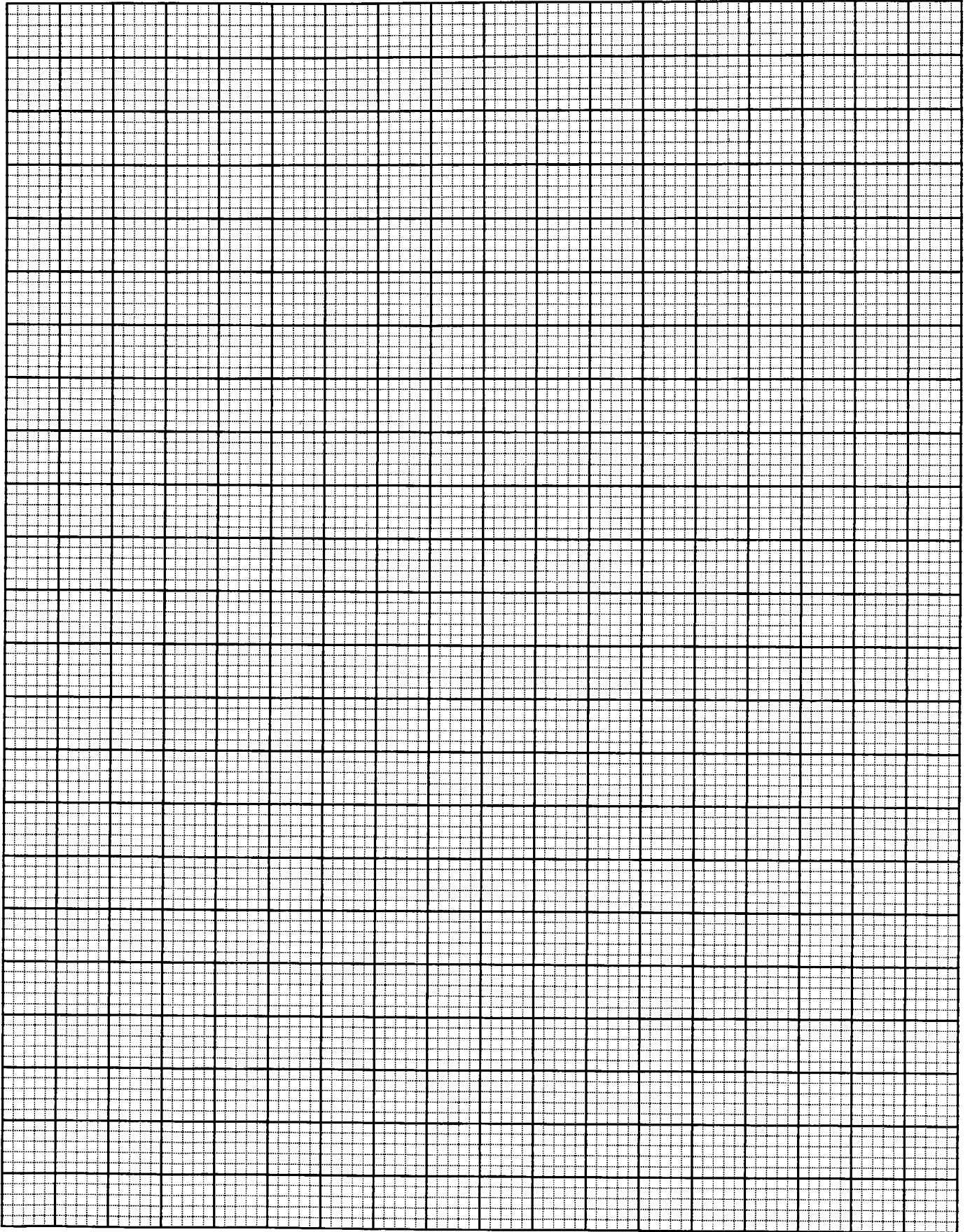
- (b) Table 2 gives data concerning the pressure, P , and volume, V , of a fixed mass of gas as it expands in a cylinder. The temperature of the gas is fixed at 300 K.

TABLE 2

Pressure, $P/10^5 \text{ Pa}$	Volume, V/m^3
4.00	0.001
2.00	0.002
1.35	0.003
1.00	0.004
0.80	0.005

- (i) On the grid provided on page 11, plot a graph of pressure, P , against volume, V .
[4 marks]
- (ii) Use your graph to determine the work done by the gas during the expansion.
[3 marks]

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- (iii) Write the ideal gas equation and use it to determine the number of moles of gas present.

[3 marks]

- (iv) Determine the change in internal energy of the gas.

[1 mark]

- (v) Determine the heat supplied to the gas during the expansion.

[2 marks]

Total 15 marks

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SECTION B

Answer ALL questions in this section.

Write your answers in the spaces provided at the end of each question.

4. (a) State Newton's 2nd law of motion. **[1 mark]**
- (b) A rock of density 2500 kg m^{-3} with volume 0.002 m^3 falls from rest in a viscous medium. The rock is acted on by a net constant downward force of 30 N (a combination of the gravitational force and a buoyant force F_b) and a viscous retarding force F_d , proportional to its velocity, v : $F_d = 5v$.
- (i) Draw a free body diagram showing the forces acting on the rock.
- (ii) Explain what is meant by a 'viscous medium'. **[2 marks]**
- (c) Calculate the
- (i) mass of the rock
- (ii) buoyant force, F_b
- (iii) initial acceleration of the rock
- (iv) terminal velocity of the rock. **[10 marks]**
- (d) Sketch graphs to show how the (i) velocity and (ii) acceleration of the rock vary with time. **[2 marks]**

Total 15 marks

Write your answer to Question 4 here.

Space for diagram.

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5. (a) (i) Differentiate between the terms ‘diffraction’ and ‘refraction’.
- (ii) Describe how the width of the aperture through which a wave is passing affects the diffraction of the wave.
- (iii) Explain how a diffraction grating produces a spectrum. **[6 marks]**
- (b) A diffraction grating having 4.55×10^5 lines per metre is used to examine the light from a glowing gas. The first order violet light emerges at an angle of 11.8° and the first order red light at an angle of 15.8° .
- Calculate the wavelengths of the two colours. **[5 marks]**
- (c) With the aid of calculations, determine what would be observed at an angle of 54.8° . **[4 marks]**

Total 15 marks

Write your answer to Question 5 here.

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6. (a) (i) Describe how a physical property of a substance which varies with temperature may be used for the measurement of temperature on an empirical centigrade scale.
- (ii) Explain how the absolute thermodynamic scale of temperature differs from that described above.
- (iii) Mercury expands only a small amount when it is heated. There are many liquids which expand a great deal more. State ONE reason why mercury is still often used in thermometers. **[6 marks]**
- (b) A resistance thermometer is placed in a bath of liquid at 0°C and its resistance is found to be $3750\ \Omega$. At 100°C , its resistance is $215\ \Omega$. The bath is cooled until the resistance is $950\ \Omega$.
- (i) Determine the new temperature of the bath, as measured using the resistance thermometer.
- (ii) The reading taken at the same time on a mercury-in-glass thermometer placed in the bath is 76°C . Suggest ONE reason for the difference between this reading and the value calculated above in b (i). **[3 marks]**
- (c) Figure 3 shows an electric water heater designed to provide a continuous supply of hot water. Water is flowing at the rate of $0.300\ \text{kg min}^{-1}$, the inlet thermometer registers 20.0°C , the voltmeter reads $120\ \text{V}$ and the ammeter reads $10.0\ \text{A}$.
- (i) Calculate the power developed by the heater.
- (ii) When a steady state is finally reached, what is the reading on the outlet thermometer?
- (iii) State ONE reason why it is NOT necessary to take into account the heat capacity of the apparatus itself. **[6 marks]**

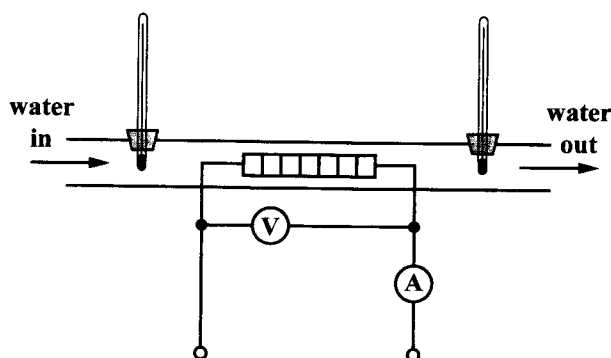


Figure 3

Total 15 marks

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END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.